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The canonical collection is used to generate which of the following parsing tables?

A. Production and symbol tables
B. First and follow tables
C. Closure and reduction tables
D. Action and goto tables

What does the term "canonical collection" refer to in LR parsing?

A. A set of LR items representing the states of the parser during parsing.
B. A table used for storing parsing actions.
C. The final output of the parsing process.
D. A list of production rules in a grammar.

In LR grammar, the R stands for _____.

A. Right shift
B. Right to left scan
C. Right most derivation
D. Right hand side of grammar

The NFA of LR(0) items represents transitions between:

A. LR(1) items.
B. LR(0) items without considering lookahead symbols.
C. Grammar symbols and their derivations.
D. Non-terminal symbols and their productions.

In the grammar "T → a b c d | a b d c", the _____ is a common prefix for non-terminal 'T'.

A. c d
B. a b
C. b d
D. a c

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In the LR table, how are the entries other than the state containing [S' → •S, $] initialized?

A. Marked as "accept"
B. Assigned with reduce actions
C. Marked as "error"
D. Assigned with shift actions

Left factoring is primarily used in which type of parsing?

A. Predictive parsing
B. Bottom-up parsing
C. Shift-Reduce parsing
D. LR parsing

In the handle representation <A → α, k>, what does A represent?

A. The non-terminal on the left-hand side of the production rule.
B. The last symbol in the input string.
C. The position of the handle within the input string.
D. The lookahead symbol.

The output from the algorithm for the construction of collection of canonical sets of LR(1) items is _____.

A. CC sets of LR(1)
B. Original Grammar G
C. CC sets of LR(0)
D. Augmented Grammar G'

Which of the following is NOT a step in the closure procedure for an LR item?

A. Add the item to the closure set.
B. Identify the symbols immediately following the dot in the item.
C. Compute the FIRST set for the nonterminal symbol following the dot.

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D. Determine the lookahead symbols for the item.

The augmented grammar has its own start symbol, while original grammar includes an additional production rule, called the augmented start production.

A. true
B. false

In Shift-Reduce parser, the handle-recognizing DFA is basically represented as a 2-dimensional table, where _____ correspond to DFA states.

A. Cells
B. Rows
C. Captions
D. Columns

The Upper Frontier is associated with which type of parsing algorithm?

A. Top-Down
B. Bottom-Down
C. Top-Up
D. Bottom-Up

The upper frontier in a shift-reduce parser is the set of symbols at the top of the parser's stack that have not been reduced yet.

A. true
B. false

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The input to the algorithm for the construction of collection of canonical sets of LR(1) items is _____.

A. Original Grammar G
B. CC sets of LR(1)
C. CC sets of LR(0)
D. Augmented Grammar G'

What is the upper frontier in a partially constructed parse tree?

A. The bottommost part of the tree.
B. The leftmost part of the tree.
C. The topmost part of the tree.
D. The rightmost part of the tree.

In the context of LR(1) parser, which of the following forms the GoTo table in the DFA representation table?

A. Columns with non-terminals
B. Columns with terminals
C. Rows with non-terminals
D. Rows with terminals

LR(0) parsing is a more powerful parsing technique than LR(1) parsing.

A. true
B. false

LR parsers are commonly used for:

A. Syntax analysis
B. Register Allocation
C. Code generation
D. Lexical analysis

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If closure(I) is "I0 = [S' → ·S, S → ·s]" for a typical grammar, then goto(I0, s) would be _____.

A. S' → S·
B. S → S·
C. S' → s·
D. S → s·

In the context of canonical collection of LR(1) items, if a terminal appears after the placeholder "·" then no further closure is possible.

A. true
B. false

The purpose of the added ε-transitions in the NFA is to:

A. Resolve conflicts in the parsing process.
B. Recognize the symbols β before a reduction by X → β.
C. Push the non-terminal X onto the parse stack.
D. Identify the start state of the grammar.

In the context of canonical collection of LR(1) items, if a non-terminal appears after the placeholder "·" then no further closure is possible.

A. true
B. false

The canonical collection is constructed by repeatedly applying which operations?

A. Closure and shift
B. Shift and reduce
C. Reduce and goto
D. Closure and goto

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The augmented grammar is particularly important in which parsing algorithm?

A. Top-down parsing.
B. Recursive descent parsing.
C. LR(1) parsing.
D. LL(1) parsing.

What does the GoTo table in the DFA representation table represent?

A. Transitioning to terminals
B. Accepting the input
C. Applying production rules
D. Transitioning to non-terminals

A top-bottom parser operates by repeatedly locating handles on the frontier of the partial parse tree and performing reductions that they specify.

A. true
B. false

In the context of canonical collection of LR(1) items, compute the state I0 for the grammar " S → E, E → E + (E) | int".

A. s = { [S → •E,],[E→·E+(E),] , [E → •int, $] }
B. s = { [E → •S,],[E→·E+(E),] , [E → •int, $] }
C. s = { [E → •S,],[S→·S+(S),] , [E → •int, $] }
D. s = { [S → •E,],[S→·S+(S),] , [E → •int, $] }

In Shift-Reduce parsing, every legal input eventually reduces the entire frontier to grammar's goal symbol.

A. true
B. false

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One of the rationales behind the Closure procedure is to:

A. Handle conflicts arising from ambiguous grammar rules.
B. Optimize the performance of the parsing algorithm.
C. Determine potential reductions and expansions in the grammar.
D. Identify grammar errors in the production rules.

In the handle representation <A → α, k>, what does k represent?

A. The index of the first symbol in the handle.
B. The number of non-terminal symbols in the handle.
C. The position of the handle within the input string.
D. The length of the handle.

If [X → α • β, c] is an LR(1) item, then "c" is a/an _____.

A. Production
B. Look-ahead symbol
C. Non-terminal
D. Terminal

In LR(1) parser, the Action table specifies the next state, while the GoTo table determines the particular action (i.e., shift / reduce).

A. true
B. false

In the reduce action, the shift-reduce parser applies an inverse production at the right end of the _____ string.

A. Bottom
B. Top
C. Right
D. left

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LR parsers CAN NOT handle left-recursive grammars.

A. true

B. false